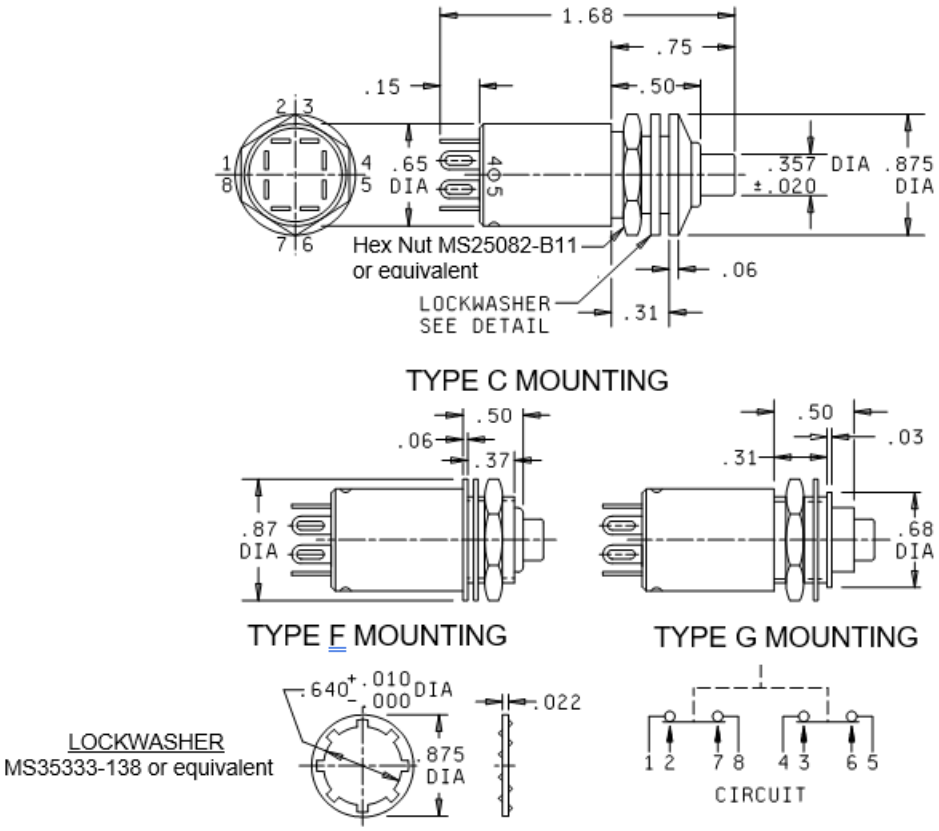
INCH-POUND
MIL-PRF-8805/38J
7 June 2023
SUPERSEDING
MIL-PRF-8805/38H
6 December 2012

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSH, MOMENTARY, 4 CIRCUIT, 4 AMPERES OR LOW LEVEL, DUSTTIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-PRF-8805.



NOTES:
1. Dimensions are in inches.
2. Unless otherwise stated, tolerances are ±.02 for two place decimals and ±.005 for three place decimals.
3. The circuit and terminal arrangement shall be permanently and legibly marked on the switch case.
4. Metric equivalents are given for general information only.
5. Mounting hardware has .625-24 UNEF02A external thread and is removable for installation.
6. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for hardware material
7. MS25082-B11 per MS25082. MS35333-138 per NASM35333.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 2 (dusttight).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: (50 g's sine wave), test condition A, MIL-STD-202-213.

Sinusoidal vibration grade: 1 (10 to 500 Hz).

Weight: 0.40 pound maximum.

Operating characteristics:

Total plunger travel: .085 ±.015 inch.

Operating force: 4 ±1 pounds.

Strength of actuator means: 25 pounds.

Terminal strength: 5 pounds.

Finish: Shall be corrosion resistant, black and lusterless.

Dielectric withstanding voltage:

Sea level: 1,000 V rms; 1,000 V rms between all terminals and mounting plate and 200 V rms between all terminals following electrical endurance.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

Sand and dust: Applicable.

Low level circuit: Applicable when specified (see table III).

Electronic logic circuit (5.0 V dc, .010 ampere): Applicable when specified (see table III); 25,000 cycles at an actuation rate of 120 cycles per minute maximum with no "stick", or "misses", allowed when tested in accordance with ECA RS448, method 17 as follows:

Test condition:

Each switch contact shall be tested using a 5.0 ±0.5 V dc, 10 ±1 mA resistive load. During each closure of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact static closure. The switch contacts need not be monitored until 10 milliseconds after the initial contact closure to exclude any contact bounce. During each opening of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact opening.

A voltage of 2.1 volts or greater across the switch terminals shall constitute a contact "miss" (failure to properly close the circuit). A voltage drop of less than 90 percent of the open-circuit voltage shall constitute a contact "stick" (failure to properly open the circuit).

The monitoring device shall either record the number of contact closures at which "sticks" and/or "misses" occur, or discontinue the test when "sticks" and/or "misses" occur.

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Operating temperature:

a. 25 percent for the test cycle at -55°C specified.

b. 25 percent for the test cycle at room ambient conditions.

c. 50 percent for the test cycle at +85°C.

Qualification:

Group submission: See table II.

Part number: See table III.

TABLE I. Electrical ratings, power circuit switches (MS27903-1 through MS27903-6).

	Sea level	
Load	28 V dc	115 V ac, 400 Hz
	(Amperes)	(Amperes)
Resistive	4	4
Inductive	2	2
Motor	2	2

NOTE: MS27903 was superseded by MIL-PRF-8805/38; the MS27903 Part Numbers (PINS) were retained.

TABLE II. Qualification inspection.

Examination or test	Basic switch	Other switch samples	Extent of Approval	
Qualification inspection table of MIL-PRF-8805	MS27903-1 or MS27903-2 (all sample units)	MS27903-3, MS27903-4, MS27903-5, MS27903-6 (2 sample units each) Visual and mechanical examination	MS27903-1 through MS27903-06	
Group I and group X of the qualification inspection table of MIL-PRF-8805 plus 2 additional samples shall be subjected to logic level circuit and operating characteristics	MS27903-7 or MS27903-8			All

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TABLE III. Part numbers and characteristics.

Part number	Type	Button	Electrical ratings
MS27903-1	C	Black	Power circuit rating (see table I)
MS27903-2	C	Red	
MS27903-3	G	Black	
MS27903-4	G	Red	
MS27903-5	F	Black	
MS27903-6	F	Red	
MS27903-7	C	Black	Low level and logic level rating
MS27903-8	C	Red	
MS27903-9	G	Black	
MS27903-10	G	Red	
MS27903-11	F	Black	
MS27903-12	F	Red	

NOTE: MS27903 was superseded by MIL-PRF-8805/38; the MS27903 Part Numbers (PINS) were retained.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced Documents:

MIL-PRF-8805
MS25082
MIL-STD-202-213
ECA RS448
NASM35333

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2023-032

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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